

Vidhi Mistry

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EDUCATION

University of Illinois Urbana-Champaign

Master of Engineering in Chemical Engineering Leadership | GPA: 3.86

Illinois, USA

Aug 2025 – Present

Nirma University

Bachelor of Technology in Chemical Engineering | GPA: 3.24

Ahmedabad, Gujarat, India

Oct 2021 – May 2025

EXPERIENCE

University of Illinois Urbana-Champaign

Graduate Research Assistant

Urbana-Champaign, IL

May 2026 – Present

- Assisted in developing machine learning models to identify safer, biodegradable alternatives to PFAS by predicting molecular toxicity, reducing the experimental effort required for chemical screening.
- Curated molecular datasets and evaluated toxicity predictions to guide experimental screening.

University of Illinois Urbana-Champaign

Graduate Research Assistant, RIPE Project

Urbana-Champaign, IL

Jun 2026 – Present

- Authored 8+ Standard Operating Procedures and safety documentation covering autoclave operation, equipment sterilization and experimental procedures, standardizing laboratory practice across the group.
- Worked in a cleanroom environment, following contamination-control and laboratory safety protocols throughout experimental work.
- Coordinated sample labeling, tracking and experimental documentation across multiple research groups and treatment conditions.

Gujarat Narmada Valley Fertilizers & Chemicals Ltd

Process Engineering Trainee

Bharuch, Gujarat, India

May 2024 – Jun 2024

- Documented Standard Operating Procedures for plant shutdown, equipment isolation, depressurization, purging and lockout-tagout, and observed a scheduled maintenance shutdown of an operating plant.
- Monitored operating parameters including temperature, pressure and flow using Distributed Control Systems under engineering supervision.
- Applied learning-from-incident findings when documenting Standard Operating Procedures, with exposure to HAZOP preparation in an operating plant.
- Evaluated large-scale syngas production units, including reformers, catalysts, heat exchangers and energy recovery systems, to understand process performance and operational efficiency.
- Ensured compliance with industrial safety standards through PPE adherence, hazard identification and safe work practices in the plant.
- Collaborated with engineering and operations teams to troubleshoot deviations and improve reliability.

Nirma University

Researcher

Ahmedabad, Gujarat, India

Dec 2024 – May 2025

"Employment of Industrial Waste Material: Calcium Carbide Residue (CCR) as a Photocatalyst for Green Hydrogen Production".

- Filed a patent application (pending) for a novel method using industrial waste Calcium Carbide Residue as a photocatalyst, achieving approximately 15% higher hydrogen generation efficiency than electrolysis alone by combining electrolysis with photocatalysis.
- Produced hydrogen at an applied potential of 1.3 V, below the 1.6-2.2 V practical operating range of PEM and alkaline electrolyzers, and quantified hydrogen yield by water displacement collected at the cathode.
- Designed a custom photo-electrocatalytic hydrogen production cell in AutoCAD and had it externally fabricated for experimental use.

Nirma University

Researcher

Ahmedabad, Gujarat, India

Jul 2023 – Sep 2023

"Wastewater Treatment of Dye Intermediate Industry for Zero-Liquid Discharge".

- Performed 15+ lab-scale experiments evaluating coagulant and flocculant combinations for dye-intermediate industry wastewater.
- Measured Chemical Oxygen Demand by the closed reflux colorimetric method and Biological Oxygen Demand by the dilution method.
- Reduced Chemical Oxygen Demand from the thousands of mg/L to approximately 100 mg/L and Biological Oxygen Demand to approximately 30 mg/L through optimized chemical dosing.
- Developed and applied modified chitosan as a biodegradable flocculant.

RESEARCH & PROJECTS

University of Illinois Urbana-Champaign

Graduate Research Project

Urbana-Champaign, IL

Feb 2026 – May 2026

"AI-Assisted Physics-Guided Modelling for Carbon Capture and Gas Separation in Polymer Membranes"

- Developed predictive machine learning models, Random Forest as baseline and XGBoost as the main model, combining Morgan fingerprints with physics-based gas transport and polymer descriptors to evaluate membrane selectivity across three gas pairs for carbon capture.
- Implemented GroupShuffleSplit and 5-fold GroupKFold validation grouped by polymer SMILES to ensure no polymer appearing in training was present in the test set.
- Validated model generalization on synthetic polymer datasets unseen by the model.

TECHNICAL SKILLS

Process & Plant: Steam reforming and syngas production (plant exposure), Heat exchange and energy recovery (plant exposure), Water electrolysis and photo-electrocatalysis, Membrane gas separation (computational), Coagulation-flocculation, Zero-liquid discharge

Operations & Safety: DCS monitoring (plant exposure), Lockout-tagout, isolation, depressurization and purging, PPE and hazard identification, HAZOP and HAZID (coursework and internship exposure), SOP development, Six Sigma Green Belt (LinkedIn Learning)

Analysis & Software: COD and BOD analysis, Chemical dosing optimization, AutoCAD, ChemCAD (coursework), MATLAB (coursework), PLC Programming (coursework), Python, R, Excel